

Water Physics 2: Electrolytes

Vincent Démery

October 3, 2026

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1 Dielectric properties of water

1.1 A few numbers on pure and sea water

The water molecule contains a central oxygen atom linked to two hydrogen atoms at a distance of 96 pm; the HOH angle is around 104° . This anisotropic shape results in a permanent electric dipole $p_0 \simeq 1.85 \text{ D} = 6 \times 10^{-30} \text{ C m}$. The weight of a water molecule is $m_{\text{H}_2\text{O}} = 18 \text{ g mol}^{-1}$.

Here, we will give the different quantities in *normal conditions*: “room temperature” $T = 20^\circ\text{C} = 293 \text{ K}$ and ambient pressure $P = 1 \text{ atm} = 1 \times 10^5 \text{ Pa}$. Under these conditions, the water density is $\rho \simeq 1 \times 10^3 \text{ kg m}^{-3}$, so that the volume occupied by a molecule is

$$v_{\text{H}_2\text{O}} \simeq \frac{m_{\text{H}_2\text{O}}}{\rho} \simeq (3.1 \text{ \AA})^3. \quad (1)$$

We will study the interplay between water and ions, or salt, and charged interfaces. The most common salt is sodium chloride, $\text{NaCl} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$. Its mass is $m_{\text{NaCl}} = 58 \text{ g mol}^{-1}$ (the mass numbers are ^{23}Na and ^{35}Cl). The salt concentration in sea water is $c \simeq 35 \text{ g L}^{-1} \simeq 0.6 \text{ mol L}^{-1}$ and in physiological conditions it is $c \simeq 9 \text{ g L}^{-1} \simeq 0.15 \text{ mol L}^{-1}$.

1.2 Water molecule and polarizability

In absence of the external field, the orientation $\mathbf{n}$ of an individual water molecule is random. Because of its dipole moment, it reorients in an external electric field $\mathbf{E}$, and acquires an average polarization $\langle \mathbf{p} \rangle = p_0 \langle \mathbf{n} \rangle$, an effect that we will compute later. For a weak external field, the average polarization is proportional to the electric field,

$$\mathbf{p} = \alpha \epsilon_0 \mathbf{E}, \quad (2)$$

where ϵ_0 is the vacuum permittivity and α is the *polarizability* of the molecule. Beyond reorientation, a water molecule may acquire a polarization through other mechanisms, such as the polarization of its electronic cloud or its deformation. In the following, we will discuss the orientation induced polarization, which is dominant, in more details.

In a medium containing a number density n of molecules with individual polarization $\mathbf{p}$, we may define the polarization density $\mathbf{P} = n\mathbf{p}$. The polarizability of a continuous medium is characterized by its susceptibility χ such that

$$\mathbf{P} = n\alpha\epsilon_0\mathbf{E} = \epsilon_0\chi\mathbf{E}. \quad (3)$$

The polarization density $\mathbf{P}(\mathbf{r}, t)$ of a medium is due to the displacement of bound charges; we denote $\rho_b(\mathbf{r}, t)$ their density, which is related to the polarization density through

$$\rho_b = -\nabla \cdot \mathbf{P}. \quad (4)$$

Since the density of bound charges is a conserved quantity, its evolution follows a conservation equation

$$\partial_t \rho_b = -\nabla \cdot \mathbf{j}_b, \quad (5)$$

where $\mathbf{j}_b(\mathbf{r}, t)$ is the current of bound charges. Taking the time derivative of Eq. (4) and using the definition of the current of bound charges, we obtain the relation between the bound current and the polarization density:

$$\mathbf{j}_b = \partial_t \mathbf{P}. \quad (6)$$

1.3 Debye relaxation model

We now focus on the polarization related to the orientation of the water molecule by the electric field and introduce the Debye model [1]. We treat the molecule as if it were far from its siblings, which is clearly not true in water. However, the physical mechanism and the orders of magnitude are correct, and the results that we obtain describe the response of water very well if effective quantities are taken.

We set the coordinates so that the electric field is oriented along $\mathbf{u}_z$: $\mathbf{E} = E\mathbf{u}_z$. First, we determine the average polarization $\langle \mathbf{p} \rangle = p_0 \langle \mathbf{n} \rangle$ as a function of the electric field and the thermal agitation. Then we will determine how the orientation responds to a time varying electric field.

Average polarization in a permanent electric field. We denote $f(\mathbf{n})$ the distribution of the orientation of the water molecule or, more precisely, its dipole. In absence of electric field, it is simply the uniform distribution, $f(\mathbf{n}) = 1/(4\pi)$, where 4π is the area of the unit sphere. With the electric field, the energy of the dipole is

$$U = -\mathbf{p} \cdot \mathbf{E} = -p_0 \mathbf{n} \cdot \mathbf{E} = -p_0 E \cos(\theta), \quad (7)$$

where θ is the orientation in spherical coordinates, such that $\mathbf{n} \cdot \mathbf{u}_z = \cos(\theta)$. In the canonical ensemble, the distribution of the orientation is

$$f(\mathbf{n}) = \frac{1}{Z} \exp\left(-\frac{U(\mathbf{n})}{k_B T}\right) = \frac{1}{Z} \exp\left(\frac{p_0 E}{k_B T} \cos(\theta)\right), \quad (8)$$

where Z is a normalization factor, defined by

$$\int f(\mathbf{n}) d\mathbf{n} = 1. \quad (9)$$

To simplify the notations, we introduce the dimensionless electric field

$$\mathcal{E} = \frac{p_0 E}{k_B T}. \quad (10)$$

Using the fact that the distribution does not depend on the spherical coordinate φ , we can integrate over the orientation easily:

$$1 = \int f(\mathbf{n}) d\mathbf{n} = \int_0^\pi f(\theta) 2\pi \sin(\theta) d\theta \quad (11)$$

$$= \frac{2\pi}{Z} \int_0^\pi \exp(\mathcal{E} \cos(\theta)) \sin(\theta) d\theta \quad (12)$$

$$= \frac{2\pi}{Z} \int_{-1}^1 \exp(\mathcal{E} u) du \quad (13)$$

$$= \frac{2\pi}{Z} \frac{e^\mathcal{E} - e^{-\mathcal{E}}}{\mathcal{E}} \quad (14)$$

$$= \frac{4\pi}{Z} \frac{\sinh(\mathcal{E})}{\mathcal{E}}. \quad (15)$$

We have changed variable to $u = \cos(\theta)$, $du = \sin(\theta) d\theta$. The normalization factor is

$$Z = 4\pi \frac{\sinh(\mathcal{E})}{\mathcal{E}} = 4\pi \left(1 + \frac{\mathcal{E}^2}{6} + \mathcal{O}(\mathcal{E}^4)\right). \quad (16)$$

The average orientation is defined by

$$\langle \mathbf{n} \rangle = \int \mathbf{n} f(\mathbf{n}) d\mathbf{n}. \quad (17)$$

It should be in the same direction as the field and we just have to compute its magnitude, that is, the average of $n_z = \cos(\theta)$. It is given by

$$\langle \cos(\theta) \rangle = \frac{2\pi}{Z} \int_0^\pi \cos(\theta) \exp(\mathcal{E} \cos(\theta)) \sin(\theta) d\theta. \quad (18)$$

It can be performed explicitly, or one can notice from Eq. (12) that it is

$$\langle \cos(\theta) \rangle = \frac{\partial \mathcal{E} Z}{Z} = \partial_{\mathcal{E}} \ln(Z) = \coth(\mathcal{E}) - \frac{1}{\mathcal{E}} = \frac{\mathcal{E}}{3} - \frac{\mathcal{E}^3}{45} + \mathcal{O}(\mathcal{E}^5). \quad (19)$$

At low electric fields, $\mathcal{E} \ll 1$, a water molecule responds linearly to the field and

$$\langle \mathbf{p} \rangle \simeq \frac{p_0 \mathcal{E}}{3} = \frac{p_0^2 E}{3k_B T}, \quad (20)$$

corresponding to a polarizability

$$\alpha = \frac{p_0^2}{3\epsilon_0 k_B T}, \quad (21)$$

and a susceptibility

$$\chi = n\alpha = \frac{np_0^2}{3\epsilon_0 k_B T}. \quad (22)$$

At large electric fields, the molecule lies along $\mathbf{u}_z$ and its polarization saturates to p_0 .

We now compute the orders of magnitude. With $p_0 \simeq 6 \times 10^{-30} \text{ C m}$, $\epsilon_0 \simeq 8.9 \times 10^{-12} \text{ F m}^{-1}$, $k_B T \simeq 4.0 \times 10^{-21} \text{ J}$ at $T = 20^\circ \text{C}$, and density $n \simeq (3.1 \times 10^{-10} \text{ m})^{-3}$, we get for the susceptibility

$$\chi \simeq 11, \quad (23)$$

which is roughly of the same order of magnitude as the value $\chi \simeq 72$ fitted on the experiments [2]. The difference is explained by the strong correlations between neighbors. Fitting the experimental data also gives a residual susceptibility $\chi' \simeq 3.73$ that is not explained by the water reorientation and is attributed to the polarization of the electronic cloud and the vibrational modes of the molecule. Because $\chi' \ll \chi$, the large static susceptibility of water is mainly due to the molecular orientation.

The critical electric field E_c to orient a water molecule corresponds to $\mathcal{E} = 1$, that is

$$E_c = \frac{k_B T}{p_0} \simeq 7 \times 10^8 \text{ V m}^{-1} \approx 1 \text{ V nm}^{-1}. \quad (24)$$

This is a very strong field, but it can be encountered close to ions or charged interfaces.

Relaxation in a time varying electric field. We now consider a time varying electric field $\mathbf{E}(t) = E_0 \exp(i\omega t) \mathbf{u}_z$ and focus on the linear response. We have to compute the time dependent orientation distribution $f(\mathbf{n}, t)$. At first order in the field and for a permanent field, the orientation distribution is

$$f(\theta) \simeq \frac{1}{4\pi} [1 + \mathcal{E} \cos(\theta)]. \quad (25)$$

We retain this shape for a time varying field and look for a response of the form

$$f(\theta, t) = \frac{1}{4\pi} [1 + a(t) \cos(\theta)]. \quad (26)$$

From the static result (19), we know that the average orientation is

$$\langle \cos(\theta(t)) \rangle = \frac{a(t)}{3}. \quad (27)$$

To determine the time dependent response, we need an evolution equation for $f(\theta, t)$. If inertia can be neglected, which is true in a liquid but not in a gas, the distribution follows the Smoluchovski equation

$$\partial_t f(\mathbf{n}, t) = k_B T \mu_r \nabla^2 f(\mathbf{n}, t) + \mu_r \nabla \cdot [f(\mathbf{n}, t) \nabla U(\mathbf{n}, t)]. \quad (28)$$

We have introduced the rotational mobility μ_r , which encodes the response of the orientation to an external force. It is related to the rotational diffusion coefficient D_r through the Einstein relation $D_r = k_B T \mu_r$. The first term describes the rotation due to thermal noise, and the second the effect of the external field, where the energy is given by Eq. (7). The differential operators have to be taken on the sphere.

For the second term on the right hand side, since U is of order 1 in the field, we can assume that $f(\theta, t) \simeq f_0 = 1/(4\pi)$, so that this term reduces to $f_0 \nabla^2 U(\mathbf{n}, t)$. We now plug in the form (26); we obtain

$$\dot{a}(t) \cos(\theta) = D_r [\nabla^2 \cos(\theta)](t) - D_r \mathcal{E}_0 [\nabla^2 \cos(\theta)] e^{i\omega t}. \quad (29)$$

Using the laplacian in spherical coordinates, we have

$$\nabla^2 \cos(\theta) = \frac{1}{\sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial \cos(\theta)}{\partial \theta} \right) = -2 \cos(\theta). \quad (30)$$

We thus end up with

$$\dot{a}(t) = 2D_r [\mathcal{E}_0 e^{i\omega t} - a(t)]. \quad (31)$$

Looking for a solution under the form $a(t) = a_0 \exp(i\omega t)$, we find

$$a_0 = \frac{\mathcal{E}_0}{1 + i\tau_D \omega}, \quad (32)$$

where we have defined the Debye relaxation time $\tau_D = (2D_r)^{-1}$.

Following the same steps as in the static case, we obtain the frequency-dependent susceptibility

$$\chi(\omega) = \frac{\chi_0}{1 + i\tau_D\omega} = \frac{\chi_0}{1 + \tau_D^2\omega^2} - \frac{i\chi_0\tau_D\omega}{1 + \tau_D^2\omega^2}, \quad (33)$$

where the static susceptibility is given by Eq. (22).

To obtain an order of magnitude of the Debye time, we use the Stokes-Einstein rotational diffusion coefficient of a sphere of diameter d in a fluid with viscosity η :

$$D_r = \frac{k_B T}{\pi\eta d^3}, \quad (34)$$

hence, with $\eta \simeq 0.9 \text{ mPa s}$ and $d \simeq 3.1 \text{ \AA}$, we get the relaxation time

$$\tau_D = \frac{\pi\eta d^3}{2k_B T} \simeq 10 \text{ ps}. \quad (35)$$

Experimental data fits very well with $\tau_D \simeq 7.28 \text{ ps}$ [2]. This excitation is in the microwave range (1 GHz to 100 GHz). It is remarkable that an estimate obtained by treating the water molecule as a sphere and its environment as a newtonian fluid is so close to the experimental result.

As explained above, there are other sources of polarization in water, and each comes with its characteristic time. The Debye relaxation is overdamped, so that there is no resonance, but other excitations are resonant and have a strong spectral signature that can be used to identify the compounds of a substance; this is the principle of spectroscopy. In general, microwave radiations excite the rotational modes of the molecules, infrared radiations ($3 \times 10^{11} \text{ Hz}$ to $3.8 \times 10^{14} \text{ Hz}$) excite their vibrational modes, and visible ($3.8 \times 10^{14} \text{ Hz}$ to $7.5 \times 10^{14} \text{ Hz}$) and ultraviolet excite their electronic modes. For water, there are an intermolecular stretching vibration at 5 THz, an intermolecular libration at 15 THz, and an intramolecular OH stretching vibration at 101 THz [3]. These characteristic frequencies are well above the Debye frequency $\nu_D = \tau_D^{-1} = 100 \text{ GHz}$.

1.4 Maxwell equations in a polarizable medium

Effective permittivity. We start from the original Maxwell equations for the electric field $\mathbf{E}(\mathbf{r}, t)$ and the magnetic field $\mathbf{B}(\mathbf{r}, t)$, in the presence of an electric charge $\rho(\mathbf{r}, t)$ and an electric current $\mathbf{j}(\mathbf{r}, t)$:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (36)$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (37)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (38)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \epsilon_0 \mu_0 \partial_t \mathbf{E}, \quad (39)$$

where μ_0 is the vacuum permeability.

In a polarizable medium, we may want to separate the charge density and current in a contribution of the bound charges ($\rho_b, \mathbf{j}_b$) and a contribution of the free charges ($\rho_f, \mathbf{j}_f$). We consider first the Maxwell-Gauss equation (36), which we rewrite using Eqs. (4) and (3):

$$\nabla \cdot \mathbf{E} = \frac{\rho_f + \rho_b}{\epsilon_0} = \frac{\rho_f}{\epsilon_0} - \chi \nabla \cdot \mathbf{E}. \quad (40)$$

We can rewrite this equation

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\epsilon}, \quad (41)$$

where $\epsilon = \epsilon_0 \epsilon_r$ is the permittivity, and $\epsilon_r = 1 + \chi$ is the *relative permittivity*.

We can do the same for the Maxwell-Ampère equation (39):

$$\nabla \times \mathbf{B} = \mu_0 (\mathbf{j}_f + \mathbf{j}_b) + \epsilon_0 \mu_0 \nabla \cdot \mathbf{E} \quad (42)$$

$$= \mu_0 \mathbf{j}_f + \epsilon_0 \mu_0 \chi \partial_t \mathbf{E} + \epsilon_0 \mu_0 \nabla \cdot \mathbf{E} \quad (43)$$

$$= \mu_0 \mathbf{j}_f + \epsilon \mu_0 \nabla \cdot \mathbf{E}, \quad (44)$$

where we have used the permittivity introduced previously.

We have thus shown that a polarizable medium is equivalent to a medium with an effective permittivity $(1 + \chi)\epsilon_0$. Because the permittivity enters in the denominator in the electrostatic interaction, a large permittivity means weak electrostatic interactions.

The difference between the permeability in water and in the vacuum is very weak: the relative permeability of water is $\mu_r = 0.999992$.

Electromagnetic waves. The equation of propagation of electromagnetic waves in the vacuum ($\rho = 0$, $\mathbf{j} = 0$) can be obtained by taking the curl of the Maxwell-Faraday equation (37). On one hand,

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\nabla^2 \mathbf{E}, \quad (45)$$

where we have used the Maxwell-Gauss equation (36). On the other hand, using the Maxwell-Ampère equation (39),

$$\nabla \times (\nabla \times \mathbf{E}) = -\partial_t \nabla \times \mathbf{B} = -\epsilon_0 \mu_0 \partial_t^2 \mathbf{E}. \quad (46)$$

Hence, we have

$$\partial_t^2 \mathbf{E} = c_0^2 \nabla^2 \mathbf{E}, \quad (47)$$

where we have introduced the speed of light

$$c_0 = \frac{1}{\sqrt{\epsilon_0 \mu_0}}. \quad (48)$$

The speed of light in a polarizable medium is

$$c = \frac{1}{\sqrt{\epsilon \mu_0}} = \frac{c_0}{\sqrt{\epsilon_r}} = \frac{c_0}{n}, \quad (49)$$

where n is the *refractive index* of the medium. The effective permittivity is in general a frequency-dependent complex number, and so is the refractive index. In the following, we write the complex refractive index as $\underline{n} = n - i\kappa$. It is related to the real and imaginary parts of the relative permittivity $\epsilon_r = \epsilon'_r - i\epsilon''_r = 1 + \chi' - i\chi''$ by $1 + \chi' = n^2 - \kappa^2$ and $\chi'' = 2n\kappa$.

A plane electromagnetic wave propagating in the direction x is described by the complex field amplitude

$$E(x, t) = E_0 \exp\left(i\omega \left[t - \frac{x}{c}\right]\right) = E_0 \exp\left(i\omega \left[t - c_0^{-1}(n - i\kappa)x\right]\right) = E_0 \exp(i[\omega t - nk_0 x]) \exp(-\kappa k_0 x), \quad (50)$$

where we have introduced the wavenumber in the vacuum $k_0 = \omega/c_0$. The wave is exponentially damped with a characteristic length $(\kappa k_0)^{-1}$, corresponding to the Beer-Lambert law. For this reason, κ is called the *extinction coefficient*.

The energy lost by the electromagnetic wave in the material is converted into heat: this is the principle of the microwave oven. With the values of the coefficients measured experimentally, the extinction coefficient is maximal for $\omega^* \simeq 220$ GHz and is $\kappa^* \simeq 2.82$. The extinction length can be read from the expression (50):

$$\ell = \frac{1}{\kappa k_0} = \frac{c_0}{\kappa \omega^*} \simeq 0.5 \text{ mm}. \quad (51)$$

This penetration length is too small to be used in a microwave oven. Instead, microwave ovens use $\omega_{\text{oven}} \simeq 15$ GHz (wavelength), for which $\kappa_{\text{oven}} \simeq 0.45$. The penetration depth is now

$$\ell_{\text{oven}} \simeq 4 \text{ cm}, \quad (52)$$

which warms up the food more uniformly.

Finally, note that the IAPWS provides tables for the static permittivity of water (that is, for $\omega = 0$) for a wide range of temperatures and pressures [4] and for the refractive index as a function of wavelength, temperature and density [5].

2 Ions in water and at interfaces

2.1 Individual ions in water

We have seen that the Debye relaxation is the slowest process in the response to a time-dependent electric field. The motion of ions occur on slightly longer timescales, as it takes longer for an ion to diffusive over

its size or the size of a water molecule than it takes for a water molecule to reverse its orientation. For this reason, when describing the motion and interaction of ions in water, we can describe the water as a medium with relative permittivity $\epsilon_r(\omega = 0) \simeq 80$. However, as we will discuss below, the dielectric properties of water can depend on the wavenumber k , resulting in non-local interactions.

In vacuum, the electric field generated by an ion with charge q is

$$E(r) = \frac{q}{4\pi\epsilon_0 r^2}. \quad (53)$$

We can compare this electric field to the electric field required to polarize a water molecule (Eq. (24)):

$$\frac{E(r)}{E_c} = \frac{qp_0}{4\pi\epsilon_0 k_B T r^2} = \frac{r_c^2}{r^2}, \quad (54)$$

where

$$r_c = \sqrt{\frac{qp_0}{4\pi\epsilon_0 k_B T}} \simeq 1.5 \text{ nm}. \quad (55)$$

Water molecules below r_c can be completely polarized by the ions. In practice, this happens for the first layer of water molecules around the ions, which is called the *hydration shell*. For the next layers, the effective permittivity should be taken into account and field is too weak to polarize the molecules.

The existence of the hydration shell has two main consequences. First, due to the energy (7) of a dipole in an electric field, $U = -p_0 E$, the total dipole energy of the hydration layer is

$$U_s = -\frac{zqp_0}{4\pi\epsilon_0 r^2}, \quad (56)$$

where z is the number of water molecules in the shell and r is the distance between the ion and the water molecules of the shell. To estimate it, we assume that $r \simeq d \simeq 3.1 \text{ \AA}$, the distance between water molecules in pure water, which is a reasonable approximation for sodium chloride, for instance. We obtain $U_s \simeq -5.4 \times 10^{-19} \text{ J} \simeq -130 k_B T \simeq -643 \text{ kJ mol}^{-1}$, which is very large. Note that the molar hydration energy takes into account that there are two moles of ions per mole of salt. By comparison, the dissociation energy of sodium chloride is $U_{\text{diss}} \simeq 787 \text{ kJ mol}^{-1} \simeq 6.6 \times 10^{-19} \text{ J} \simeq 160 k_B T$. According to our estimate, the hydration energy is of the same order of the dissociation energy, which explains why sodium chloride dissociates in water but not in air.

Second, the water molecules of the hydration shell are not free to rotate, and thus do not contribute to the polarization of the medium. As a consequence, the conductivity decreases with the salt concentration, and the dependence is linear at low concentration :

$$\chi(c) = \chi(c = 0) - \beta c. \quad (57)$$

We can estimate this effect. We assume that an ion cancels the effect of $z \simeq 6$ water molecules, assuming a cubic lattice. As a consequence, 1 mol of salt contains 2 mol of ions, which cancel the effect of 12 mol of water molecules. Since there are 56 mol of water in a liter, with 1 mol L⁻¹ of solvent 20 % of the water molecules do not contribute to the susceptibility, which should thus decrease by the same factor. Taking the fitted value of 72 for the susceptibility attributed to the water orientation, this gives $\beta \simeq 15 \text{ L mol}^{-1}$. Measurements give $\beta \simeq 15 \text{ L mol}^{-1}$ [6], which is not too far. The relative permittivity of sea water is thus $\epsilon_r \simeq 74$ instead of 80 for pure water. The effect is linear up to 2 mol L⁻¹, and then the effect decreases.

2.2 Interaction between ions

The potential energy of two ions with charges q and q' at a distance r in water is given by the electrostatic energy with an effective permittivity ϵ :

$$U(r) = \frac{qq'}{4\pi\epsilon r}. \quad (58)$$

Actually, due to the molecular structure of water, the potential may be more complicated at short distances. This complexity can be formulated in Fourier space [7],

$$\tilde{U}(k) = \frac{qq'}{\epsilon(k)k^2}, \quad (59)$$

where the dependence of the permittivity reflects the molecular structure of water and has been made explicit. We may expect this dependence to be significant when k^{-1} is of the order of the diameter of a

water molecule. However, this description remains linear and does not allow to describe properly effects such as the existence of an hydration shell.

The electrostatic interaction (58) is long ranged, meaning that ions interact from far away. However, a characteristic distance may be obtained by comparing the interaction energy with the thermal energy : $U(\ell_B) = k_B T$, leading to

$$\ell_B = \frac{q^2}{4\pi\epsilon k_B T} \simeq 7 \text{ \AA}. \quad (60)$$

This length is the *Bjerrum length*. It describes the distance of closest approach of ions with the same sign, or the distance below which ions of opposite sign feel an attraction stronger than the electrostatic energy.

2.3 Charged interfaces

We now consider charged solid-water interfaces. The solid can be conducting, as for an electrode, in which case a potential V is imposed and the surface charge σ adjusts, as we will see below. If the solid is an insulator, its surface can have acquire a charge if it bears ionic groups that partly dissociate in water, so that some charge is released in the liquid while the opposite charge sticks to the interface. In this case, the surface charge is fixed. An infinite planar surface creates a uniform electric field $E = \sigma/(2\epsilon)$ perpendicular to the surface.

The energy of an ion with charge q is thus linear to its distance x from the surface:

$$U(x) = -\frac{\sigma q x}{2\epsilon}. \quad (61)$$

As for the interaction between two ions, we can define a characteristic length by comparing this energy with the thermal energy: this is the *Gouy-Chapman length*

$$\ell_{GC} = \frac{2\epsilon k_B T}{\sigma q} = \frac{q}{2\pi\ell_B\sigma}. \quad (62)$$

This length is inversely proportional to the surface charge, and we get $\ell_{GC} \simeq 3.6 \text{ nm}$ for $\sigma = 1 \times 10^{-2} \text{ C m}^{-1}$.

The shape of the potential (61) seems to indicate that the effect of a charged surface does not decay with the distance, so that the whole solution is affected. The charged surface attracts oppositely charged ions from the whole solution, and repels likely charged ions to infinity. For instance, the Boltzmann probability distribution of an oppositely charged ion is $p(x) \propto \exp(-x/\ell_{GC})$. This tends to polarize the solution, which in turns tend to reduce the electric field in the solution.

2.4 Mean field description, Poisson-Boltzmann equation

To describe the effect of a charged interface on the concentration of ions in the solution, we consider a scale at which many ions interact. To be more general, we consider a solution containing several ionic species i with charge $q_i = z_i q$ and concentration n_{i0} far from the charged surface, where q is the elementary charge and the integer z_i is the valency. The solution should be electroneutral:

$$\sum_i q_i n_{i0} = q \sum_i z_i n_{i0} = 0. \quad (63)$$

We want to relate the electrostatic potential $V(\mathbf{r})$ and the ionic densities $n_i(\mathbf{r})$. On one hand, these quantities are related by the Maxwell-Gauss or Poisson equation (36), which reads for the electrostatic potential

$$\nabla^2 V(\mathbf{r}) = -\frac{q}{\epsilon} \sum_i z_i n_i(\mathbf{r}). \quad (64)$$

On the other hand, if we consider that each ion interacts independently with the electrostatic potential $V(\mathbf{r})$, its distribution is given by the Boltzmann distribution, which is proportional to $\exp(-q z_i V(\mathbf{r})/k_B T)$. The ionic densities then follow by assuming that the densities are given by n_{i0} where $V(\mathbf{r}) = 0$:

$$n_i(\mathbf{r}) = n_{i0} \exp\left(-\frac{z_i q V(\mathbf{r})}{k_B T}\right). \quad (65)$$

Combining the Poisson equation (64) and the Boltzmann equation (65), we obtain the *Poisson-Boltzmann equation* for the electrostatic potential:

$$\nabla^2 V(\mathbf{r}) = -\frac{q}{\epsilon} \sum_i z_i n_{i0} \exp\left(-\frac{z_i q V(\mathbf{r})}{k_B T}\right). \quad (66)$$

2.5 Debye-Hückel limit

The argument of the exponentials in the Poisson-Boltzmann equation (66) is small if

$$|V(\mathbf{r})| \ll V_c = \frac{k_B T}{q} \simeq 25 \text{ mV}. \quad (67)$$

Under this condition, the exponentials can be linearized. The terms that do not depend on the potential cancel due to the electroneutrality condition. We thus obtain a linear equation for the potential:

$$\nabla^2 V(\mathbf{r}) = \left(\sum_i \frac{z_i^2 q^2 n_{i0}}{\epsilon k_B T} \right) V(\mathbf{r}) \quad (68)$$

Dimensionally, the term in parentheses is the inverse of a length squared; this defines the Debye length

$$\ell_D = \left(\sum_i \frac{z_i^2 q^2 n_{i0}}{\epsilon k_B T} \right)^{-1/2}. \quad (69)$$

With a symmetric monovalent salt, there are two ionic species with valencies ± 1 and a common reference concentration n_s , so that the Debye length reduces to

$$\ell_D = \sqrt{\frac{\epsilon k_B T}{2 q^2 n_s}}. \quad (70)$$

It can also be expressed with the Bjerrum length and the salt concentration:

$$\ell_D = \frac{1}{\sqrt{8\pi \ell_B n_s}}. \quad (71)$$

Its value in a solution with 1 mol L^{-1} of salt is $\ell_D \simeq 3 \text{ \AA}$, and its value at different concentrations can be deduced using the simple concentration dependence:

$$\ell_{D,\text{sea}} \simeq 4 \text{ \AA}, \quad (72)$$

$$\ell_{D,\text{phys}} \simeq 8 \text{ \AA}. \quad (73)$$

To understand better the signification of the Poisson-Boltzmann equation, we compute the potential generated by a planar charged surface located at $x = 0$ in the half-space $x > 0$. The surface charge is denoted σ . For symmetry reasons, the potential depends only on the distance x to the charged surface. Inserting the potential $V(x)$ in the linearized Poisson-Boltzmann equation, we obtain

$$\partial_x^2 V(x) = \ell_D^{-2} V(x). \quad (74)$$

The two independent solutions of this equation are $V(x) \propto \exp(\pm x/\ell_D)$. Eliminating the divergent solution, and taking the boundary condition $V(0) = V_s$, we obtain

$$V(x) = V_s \exp(-x/\ell_D). \quad (75)$$

Last, we have to relate the surface potential to the surface charge. To do so, we compute the electric field at $x = 0^+$ from the solution (75):

$$\mathbf{E}(0^+) = -\partial_x V(0^+) \mathbf{u}_x = \frac{V_s}{\ell_D} \mathbf{u}_x. \quad (76)$$

The electric field can also be obtained by assuming that it is zero for $x < 0$ and using the Maxwell-Gauss equation (36): $\mathbf{E}(0^+) = (\sigma/\epsilon) \mathbf{u}_x$. Equating the two expressions, we get

$$V_s = \frac{\ell_D \sigma}{\epsilon} : \quad (77)$$

the surface potential depends linearly on the surface charge.

With this solution, we compute the electric charge in the solution, in the Debye-Hückel limit:

$$\sum_i \int_0^\infty z_i q n_i(x) dx = \sum_i \int_0^\infty z_i q n_{i0} \exp\left(-\frac{z_i q V(x)}{k_B T}\right) dx \quad (78)$$

$$\simeq - \sum_i \frac{z_i^2 q^2 n_{i0}}{k_B T} \int_0^\infty V(x) dx \quad (79)$$

$$= -\epsilon \ell_D^{-2} \times V_s \ell_D \quad (80)$$

$$= -\sigma, \quad (81)$$

meaning that the ions *screen* the surface charge. The ions of the opposite charge that are attracted by the surface are called *counterions*. As a consequence, its effect extends over a length ℓ_D and not over the whole solution.

The consequence of having free charges (here, ions) in a medium is thus very different of the consequence of having bound charges, as in a polarizable medium. The bound charges modify the permittivity, but they do not change the nature of the interactions: the electrostatic interaction becomes weaker, but it remains long ranged. On the contrary, free charges modify the nature of the electrostatic interaction by making it short range. That the Debye length decreases with the salt concentration means that, if there are more ions in the solution, they can screen a charge over a shorter length.

2.6 Nonlinear solution of the Poisson-Boltzmann equation

We now investigate the effect of the nonlinearity of the Poisson-Boltzmann equation, restricting ourselves to the case of a symmetric monovalent salt. The Poisson-Boltzmann equation (66) becomes

$$\nabla^2 V(\mathbf{r}) = \frac{2q n_s}{\epsilon} \sinh\left(\frac{qV(\mathbf{r})}{k_B T}\right). \quad (82)$$

Introducing the reduced potential

$$\psi(\mathbf{r}) = \frac{qV(\mathbf{r})}{k_B T} = \frac{V(\mathbf{r})}{V_c}, \quad (83)$$

and using the Debye length, it takes a simpler form:

$$\nabla^2 \psi(\mathbf{r}) = \ell_D^{-2} \sinh(\psi(\mathbf{r})). \quad (84)$$

We now write this equation in the half-space $x > 0$ bounded by a charged surface at $x = 0$, using the reduced length $w = x/\ell_D$:

$$\partial_w^2 \psi(w) = \sinh(\psi(w)). \quad (85)$$

It takes a few lines to check that the solution to this equation is

$$\tanh\left(\frac{\psi(w)}{4}\right) = \tanh\left(\frac{\psi_s}{4}\right) \exp(-w). \quad (86)$$

So that the solution as a function of x reads

$$\tanh\left(\frac{\psi(x)}{4}\right) = \tanh\left(\frac{\psi_s}{4}\right) \exp\left(-\frac{x}{\ell_D}\right). \quad (87)$$

It is immediate to check that in the limit where $|\psi(x)| \ll 1$, this solution reduces to the solution (75) of the linear equation.

We can now relate the surface potential ψ_s to the surface charge σ . As in the linear case, we should have

$$E(0^+) = \sigma/\epsilon = -\partial_x V(0^+) = -V_c \partial_x \psi(0^+). \quad (88)$$

In order to get the derivative of the potential, we differentiate the relation (87):

$$\frac{\partial_x \psi(x)}{4} \cosh\left(\frac{\psi(x)}{4}\right)^{-2} = -\ell_D^{-1} \tanh\left(\frac{\psi_s}{4}\right) \exp\left(-\frac{x}{\ell_D}\right), \quad (89)$$

where we have used that $\tanh' = \cosh^{-2}$. We thus get

$$\partial_x \psi(0^+) = -\frac{2}{\ell_D} \sinh\left(\frac{\psi_s}{2}\right). \quad (90)$$

Combining the equations above, we arrive at

$$\sinh\left(\frac{\psi_s}{2}\right) = \frac{\sigma \ell_D}{2\epsilon V_c} = \frac{\sigma \ell_D q}{2\epsilon k_B T} = \frac{\ell_D}{\ell_B} = \frac{\Sigma}{2}, \quad (91)$$

where we have defined the reduced surface charge

$$\Sigma = \frac{\sigma \ell_D q}{\epsilon k_B T}. \quad (92)$$

We can identify two regimes in the relation (91). If $\Sigma \ll 1$, the solution is

$$\psi_s = \Sigma. \quad (93)$$

With dimensional variables, it means that

$$V_s = V_c \psi_s = \frac{\sigma \ell_D}{\epsilon}; \quad (94)$$

we recover the Debye-Hückel limit. If $\Sigma \gg 1$, then we should also have $\psi_s \gg 1$, so that $\sinh(\psi_s/2) \simeq \exp(\psi_s/2)/2$, which allows us to extract ψ_s :

$$\psi_s = 2 \ln \Sigma. \quad (95)$$

In this situation, the nonlinearity of the Poisson-Boltzmann equation makes the screening is more efficient and the surface potential increases only weakly with the surface charge. We note that, in the large surface charge regime, far enough for the surface, the solution (87) can be linearized so that

$$\psi(x) = 4 \tanh\left(\frac{\psi_s}{4}\right) \exp\left(-\frac{x}{\ell_D}\right). \quad (96)$$

We recover the solution when the surface charge is small, Eq. (75), for an effective surface potential

$$\psi_{s,\text{eff}} = \Sigma_{\text{eff}} = 4 \tanh\left(\frac{\psi_s}{4}\right) = 4 \frac{\Sigma - 1}{\Sigma + 1}, \quad (97)$$

so that $\Sigma_{\text{eff}} \rightarrow 4$ when $\Sigma \rightarrow \infty$, meaning that no surface charge appears larger than $\Sigma_{\text{eff}} = 4$ at any distance from the surface. Another way to see this effect is that from the non-linear solution (87), we have the inequality

$$\tanh\left(\frac{\psi(x)}{4}\right) \leq \exp\left(-\frac{x}{\ell_D}\right). \quad (98)$$

The reduced potential is bounded by a quantity that does not depend on the surface charge.

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